CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Urban traffic congestion continues to grow with vehicle ownership, and the intersection remains the most common network bottleneck. Intelligent Transportation Systems (ITS) research has therefore focused on Traffic Signal Control (TSC), moving from fixed-time and actuated controllers toward systems that read live conditions and respond accordingly, spanning fuzzy logic, metaheuristics, reinforcement learning, and generative AI. Incident management is an equally important but less-studied concern, since accidents are a major source of non-recurring congestion that most Traffic Control Centres still handle through manual observation and delayed reporting.

A recurring observation across the literature is that research effort concentrates on the control algorithm itself, while the surrounding operational system — monitoring, citizen reporting, administrator override, and performance reporting — receives little attention. This chapter reviews ten recent contributions spanning adaptive signal control, evolutionary and swarm optimisation, reinforcement learning, generative AI in ITS, and connected-vehicle control, extracting the design principles, benchmarks, and limitations relevant to UrbanFlow, a rule-based adaptive signal optimisation module combined with role-based monitoring, incident-reporting, and dashboard-visualisation, evaluated in SUMO. Deep learning, autonomous-vehicle coordination, edge computing, digital twins, and complex multi-agent RL are intentionally excluded from scope and discussed only as future extensions.

2.2 Literature Review

Each of the ten papers below is discussed for its objective, method, results, strengths/weaknesses, and relation to UrbanFlow, so the discussions feed coherently into the comparative table and gap analysis that follow.

2.2.1 Adaptive Traffic Light Systems: A Consolidated Review

Nautiyal, Gangodkar, and Diwakar [1] review ML, IoT, and fuzzy-logic approaches to Adaptive Traffic Light Systems, synthesising SUMO-based studies against Average Queue Length (AQL), Average Waiting Time (AWT), and Average Travel Time (ATT). Adaptive systems reduced AQL by up to 30%, AWT by ~25%, and improved ATT by ~20%, with a 15% throughput gain. As a synthesis it validates no single deployable system and ignores incident management or administrator control. For UrbanFlow, it provides a literature-grounded benchmark for its own congestion metrics and supports a rule-based approach, since much of the benefit comes from moving off fixed-time signalling rather than from any one technique.

2.2.2 Evolutionary and Swarm Intelligence Approaches

Bhattacharyya, Gupta, Marisha et al. [2] conduct a PRISMA-guided review of 50 studies (2015–2025) on Genetic Algorithms and Particle Swarm Optimisation for TSC. Hybrid EA-SI methods outperform single algorithms, cutting average delay by up to 28.9%, with PSO favoured for real-time use and GA for offline planning. The methods demand heavy iterative tuning unsuitable for a small deployment. UrbanFlow does not implement evolutionary optimisation; this review justifies that choice, since the tuning burden is disproportionate to a mini-project, supporting a simpler rule-based alternative.

2.2.3 Reinforcement-Learning-Based TSC: Taxonomy and Evolution

Xiao et al. [3] survey the evolution of RL-based TSC from tabular Q-learning/SARSA to modern Deep Reinforcement Learning (DRL), taxonomising control scope into single-intersection, independent, collaborative, and hierarchical learning. The field has moved toward sophisticated network-wide DRL, but these frameworks assume substantial training data and GPU-level computation unrealistic for a mini project. UrbanFlow's rule-based controller sits at the simpler, non-DRL end of this spectrum, with DRL framed as a future extension.

2.2.4 Coordination in RL and DRL for Traffic Lights

Saadi, Abghour, Chiba, Moussaid, and Ali [4] survey single- and multi-agent RL/DRL techniques (Q-Learning, DQN, Deep SARSA) for signal coordination. Critically, none of the reviewed studies had been tested on real-world traffic data at the time of writing, though MARL coordination is the field's main direction of progress. This confirms that SUMO-based simulation, as UrbanFlow adopts, is standard and accepted practice, and justifies presenting UrbanFlow's results as indicative rather than field-proven.

2.2.5 Recent Progress in RL-Based TSC

A 2025 review in Intelligent Transportation Infrastructure [5] updates RL-based TSC literature to March 2025. One representative case study found an RL controller reduced waiting time by ~16% versus static control. Its DRL-centric focus still exceeds a mini project's computational resources. For UrbanFlow, this independently confirms the order-of-magnitude benefit reported in [1] and underscores the need for realistic, locally representative SUMO demand scenarios.

2.2.6 AI in Transportation: Mapping the Research Landscape

A bibliometric meta-review in SN Computer Science [6] classifies 272 review papers by transportation mode, purpose, and aim, showing research concentrates on road traffic and autonomous-vehicle perception while integrated incident-management with public reporting remains thin. This contextualises UrbanFlow's scope — combining signal control, incident reporting, and dashboards — within a landscape that usually treats these as separate problems, supporting the research-gap analysis in Section 2.4.

2.2.7 Generative AI Applications in ITS

Rong, Ma, Ren, Lin, and Jia [7] review Generative AI (AIGC) across ITS — dialogue/reasoning, prediction, and multimodal generation — applied to demand forecasting and synthetic scenario generation. Adoption in real-time operational systems remains early and experimental. This paper delineates scope for UrbanFlow: generative AI is acknowledged only as a plausible future extension (e.g., synthetic incident scenarios), not part of the implemented feature set.

2.2.8 Citywide TD-Learning-Based Signal Control in SUMO

Reza, Ferreira, Machado, and Tavares [8] propose a modified SARSA(λ) with Gaussian eligibility-trace decay and MaxAbs scaling, combined with A* routing, evaluated on a 21-intersection SUMO network. Average waiting time fell ~59.9%, stops fell ~17.55%, and trip speed improved 3.4%. It assumes autonomous-vehicle connectivity and an elaborate RL pipeline outside mini-project scope. UrbanFlow adopts SUMO on the strength of this evidence and treats waiting-time and queue-length as its own benchmark metrics, without the underlying SARSA(λ) machinery.

2.2.9 Resilient Adaptive Control via Multi-Agent RL

Yang, Yu, Feng, and Ochieng [9] use decentralised A2C multi-agent RL, blending a resilience metric with time-to-collision safety, to improve network recovery from accident-induced disruption, reporting a 5.8% overall improvement over benchmarks. Incidents are modelled purely in simulation, not from live reporting, and training overhead is high. This is highly relevant to UrbanFlow's incident-management scope: it validates that integrating incident awareness with signal control improves performance, a principle UrbanFlow implements through simpler, human-in-the-loop Incident Reporting, Verification, and Manual Override modules.

2.2.10 Adaptive TSC in a Connected Vehicle Environment

Jing, Huang, and Chen [10] systematically review 26 studies on connected-vehicle (V2V/V2I) adaptive signal control, finding it enables more precise vehicle-level traffic-state estimation, but depends on widespread connected-vehicle penetration unrealistic in most cities, including India. This justifies UrbanFlow's reliance on conventional, non-connected sensing (Vehicle Count and Traffic Monitoring) as a realistic near-term alternative, while the review's quality-evaluation framework offers a template for structuring UrbanFlow's own evaluation.

2.2.11 Cross-Cutting Observations

Reading the ten papers together surfaces three patterns. First, there is broad consensus on evaluation metrics — queue length, waiting time, travel time, throughput, and stop count [1, 2, 5, 8, 9] — letting UrbanFlow benchmark against established indicators. Second, algorithmic sophistication trades off against deployability: the studies with the largest gains [2, 8] also carry the heaviest tuning or connectivity requirements, while those most candid about real-world validation [4, 10] flag a persistent simulation-to-reality gap; UrbanFlow sits deliberately on the deployable side. Third, incident management appears only as a simulated stress-test [9] rather than a citizen- or operator-driven workflow — the specific gap UrbanFlow is designed to fill (Section 2.4).

2.3 Comparative Analysis

Table 2.1 summarises the ten studies. Reinforcement learning, evolutionary optimisation, and multi-agent coordination report strong percentage gains but trade this for higher computational cost, larger data needs, and often unrealistic assumptions about data availability or connectivity. None of the ten studies combines signal optimisation with an integrated, user-facing incident-reporting workflow, public dashboard, and administrator override in one system — precisely the combination UrbanFlow proposes.

Table 2.1: Comparative Analysis of Reviewed Literature

Author(s)	**Method**	**Key Result**	**Relation to UrbanFlow**
Nautiyal et al. [1]	Comparative review (ML, IoT, fuzzy logic)	AQL −30%, AWT −25%, ATT −20%	Benchmarks for UrbanFlow's congestion metrics
Bhattacharyya et al. [2]	PRISMA review of GA/PSO/hybrid EA-SI	Hybrid methods cut delay by ~29%	Justifies simpler rule-based logic over heavy tuning
Xiao et al. [3]	RL/DRL taxonomy survey	Maps single- to multi-agent DRL control	Positions UrbanFlow at the simple, non-DRL end
Saadi et al. [4]	RL/DRL/MARL coordination survey	No study tested on real-world data	Validates SUMO-only evaluation as standard practice
ITI Review [5]	RL-based TSC review (2015–2025)	RL cut waiting time ~16% vs static control	Confirms benefit of even simple adaptive control
SN CompSci Meta-Review [6]	Bibliometric review of 272 papers	Incident + public reporting is under-researched	Confirms UrbanFlow's research-gap positioning
Rong et al. [7]	Generative AI (AIGC) in ITS	AIGC still experimental for live operations	Scopes generative AI out as a future extension
Reza et al. [8]	Modified SARSA(λ) + A* routing	Waiting time −59.9%, stops −17.6%	Confirms SUMO as suitable for multi-junction testing
Yang et al. [9]	Decentralised A2C multi-agent RL	+5.8% resilience under simulated accidents	Motivates incident-reporting + override module
Jing, Huang & Chen [10]	Systematic review, connected-vehicle ATSC	Needs high connected-vehicle penetration	Justifies conventional, non-connected sensing

2.4 Research Gap

The literature shows substantial progress in adaptive TSC across fuzzy/IoT [1], evolutionary/swarm [2], RL/DRL [3, 4, 5], generative AI [7], and connected-vehicle control [10], but several gaps are directly relevant to a practically deployable, college-level system:

• No study integrates a citizen/operator-driven incident-reporting and verification pipeline with signal control [9].

• DRL, multi-agent coordination, and tuned evolutionary methods [2, 3, 4, 5, 9] demand training infrastructure disproportionate to a mini project.

• Connected-vehicle [10] and autonomous-vehicle [8] approaches presuppose sensing infrastructure unavailable in most Indian intersections.

• Reviewed controllers largely operate autonomously, with little support for manual override or administrator audit.

• No reviewed paper describes a public-facing portal or dashboard alongside the control algorithm.

• As noted in [4], no surveyed coordination technique had been validated on real traffic data — an unresolved simulation-to-reality gap.

• Learning-based controllers [3, 4, 9] are demonstrated on small/synthetic networks and assume GPU resources and specialist tuning [2, 3, 4] beyond a single-semester mini project.

UrbanFlow addresses these gaps with an adaptive, SUMO-evaluated signal-optimisation module, an integrated Incident Reporting and Verification workflow, Manual Signal Override, role-based dashboards, and a lightweight approach independent of connected-vehicle infrastructure or specialised computation — demonstrating that a smaller, more transparent system can still deliver a coherent, end-to-end workflow.

2.5 Proposed System Justification

The reviewed literature consistently favours algorithmic sophistication over operational completeness. UrbanFlow is deliberately positioned at the opposite end of that trade-off. Since even comparatively simple adaptive logic produces queue-length and waiting-time gains of a similar order to more complex learning-based methods [1, 5], the marginal benefit of DRL or evolutionary optimisation is unlikely to justify the added training time and computational cost within a mini-project timeframe.

Simplicity and cost: UrbanFlow's rule-based adaptive optimisation uses real-time vehicle-count, queue-length, and waiting-time data rather than a trained network or tuned metaheuristic, avoiding the training cycles documented across [2, 3, 4]. Web-based architecture: role-based authentication across the Public User Portal, Traffic Control Centre Dashboard, and Administrator Dashboard lets the system be deployed and evaluated without proprietary infrastructure — a user-facing layer absent from the reviewed, largely simulation-only contributions.

SUMO simulation: consistent with most reviewed studies [1, 5, 8, 9], UrbanFlow measures queue length, waiting time, and throughput against an established baseline. Public reporting and override: Incident Reporting and Verification respond directly to the gap in Section 2.4, while Manual Override gives the Traffic Control Centre a human-in-the-loop safeguard absent from most automated controllers, unlike the purely simulated disruptions in [9]. Administrator control: the Administrator Dashboard, Traffic Data Storage, and Report Generation keep decisions auditable, addressing the operational-completeness gap common across the purely algorithmic literature — every signal-timing decision stays traceable, unlike opaque trained controllers. Excluded from scope: deep learning, autonomous-vehicle coordination, edge computing, digital twins, and complex multi-agent RL remain valid directions for a subsequent, full-scale project.

2.6 Summary

This chapter reviewed ten contributions spanning adaptive signal systems, evolutionary/swarm optimisation, RL/DRL, AI-in-transportation surveys, generative AI, and connected-vehicle control. The literature confirms adaptive signal control delivers substantial, well-documented reductions in queue length, waiting time, and travel time relative to fixed-time systems, while revealing a persistent gap around incident integration, manual oversight, public interaction, and real-world validation.

UrbanFlow is positioned to close this practical gap rather than compete on algorithmic complexity, combining rule-based adaptive optimisation, SUMO-based evaluation, incident reporting/verification, manual override, and role-based dashboards in one platform, while reserving deep learning, autonomous-vehicle coordination, and digital twins for future extension. Two conclusions follow: the empirical case for adaptive, data-driven control over fixed-time systems is strong and consistent across a decade of research, giving confidence in UrbanFlow's own SUMO-evaluated module; and the operational layer around that control logic — incident reporting, manual oversight, role-based access, and public visibility — remains under-served by the existing literature, which is precisely the space UrbanFlow occupies.

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